

ECE 370 Electromagnetic Fields and Waves

HOMEWORK #1

I expect you to clearly show me how you solved these problems – an orderly and visible presentation of your thoughts and process toward solution. I want to easily see that you understand what you are doing.

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1- Find the results of the following products of unit vectors. Show your work clearly.

a) $\hat{a}_\phi \cdot \hat{a}_x$

b) $\hat{a}_R \cdot \hat{a}_y$

c) $\hat{a}_r \times \hat{a}_R$

d) $\hat{a}_\theta \times \hat{a}_z$

2- Vectors $\vec{A}$ and $\vec{B}$ are given in rectangular coordinates by

$$\vec{A} = 4\hat{x} + 2\hat{y} - \hat{z}$$

$$\vec{B} = -2\hat{x} + 3\hat{z}$$

Find:

a) The scalar component, or projection, of $\vec{B}$ in the direction of $\vec{A}$.

b) The vector component of $\vec{B}$ in the direction of $\vec{A}$.

c) The vector component of $\vec{B}$ perpendicular to $\vec{A}$.

3- Transform the rectangular coordinate point $P = (1,2,3)$ into cylindrical and spherical coordinates.

Problem	Course outcome	Grade
1	1-d	/40
2	1-d	/40
3	1-d	/20
TOTAL:		/100